

Curtin University

Dubai

ISYS2014
Database Systems

USER GUIDE

FOR FURQAN_22526025 DATABASE

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Introduction

This user guide provides step-by-step instructions how to recreate my database 'Furqan_22526025', insert data into the tables, execute the SQL queries, and Python scripts that were solely developed for this assignment.

Prerequisites

before you begin, please make sure that you have the following software installed and running or your computer:

1. Linux Operating System (I used Ubuntu)
2. MY SQL Server 8.0
3. Command-Line Interface: terminal
4. Python 3 and the mysql-connector installed

    Follow the following steps for step 4:

    a. sudo apt instal python3-pip
    b. sudo apt install python3-mysql.connector

Download the Folder

The zipped folder submitted to the Black Board contains all the required files to recreate the database successfully. Please make sure that you download the folder and extract all the files in your current working directory.

Step-By-Step Implementation Guide:

1. **Database and Table Creation:** open your terminal and log in to the MySQL server as the root user using the following command:

   sudo mysql -u root -p

   After this you will be prompted for your password, once you enter it you would have successfully logged into your MySQL server.

2. **Create the Database:** please execute the following commands to create the database:

   CREATE DATABASE Furqan_22526025;

   USE Furqan_22526025;

   You can skip this step as my first script contains these commands as well.

3. **Create Tables and Constraints:** execute the first script to create the tables, primary keys, foreign keys, and other constraints:

   SOURCE create_tables.sql;

   Alternatively, you can run the commands manually too, but this is easier.

   You should see something like this on your terminal:

```
mysql> SOURCE create_tables.sql;
Query OK, 1 row affected (0.03 sec)

Database changed
Query OK, 0 rows affected (0.25 sec)

Query OK, 0 rows affected (0.12 sec)

Query OK, 0 rows affected (0.14 sec)

Query OK, 0 rows affected (0.14 sec)

Query OK, 0 rows affected (0.15 sec)

Query OK, 0 rows affected (0.17 sec)

Query OK, 0 rows affected (0.22 sec)

Query OK, 0 rows affected (0.17 sec)

Query OK, 0 rows affected (0.14 sec)
```

4. **Insert Data into the Tables:** now you have to execute the second script to insert data in the tables that were just created using the following command:

   SOURCE insert_data.sql;

After executing that you should see something like this:

```
mysql> SOURCE insert_data.sql;
Query OK, 25 rows affected (0.06 sec)
Records: 25  Duplicates: 0  Warnings: 0

Query OK, 50 rows affected (0.05 sec)
Records: 50  Duplicates: 0  Warnings: 0

Query OK, 18 rows affected (0.03 sec)
Records: 18  Duplicates: 0  Warnings: 0

Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

Query OK, 25 rows affected (0.05 sec)
Records: 25  Duplicates: 0  Warnings: 0

Query OK, 69 rows affected (0.04 sec)
Records: 69  Duplicates: 0  Warnings: 0

Query OK, 10 rows affected (0.06 sec)
Records: 10  Duplicates: 0  Warnings: 0

Query OK, 8 rows affected (0.04 sec)
Records: 8  Duplicates: 0  Warnings: 0

Query OK, 11 rows affected (0.04 sec)
Records: 11  Duplicates: 0  Warnings: 0
```

5. **Verification**: To verify tables were created and the data has been inserted correctly you can use the following commands:

   SHOW TABLES;

   DESC <Table_name>;

   (use ‘pager less -SFX’ so the output fits the terminal)

   SELECT * FROM <Table_name>;

6. **Create Advanced Features**: execute the third script to create the stored procedures, triggers, indexes, and views using the following command:

   SOURCE advanced_features.sql;

You should see the OK Queries like before.

7. **Testing Advanced Features:** now to run the following scripts to test the advanced features using the following commands:

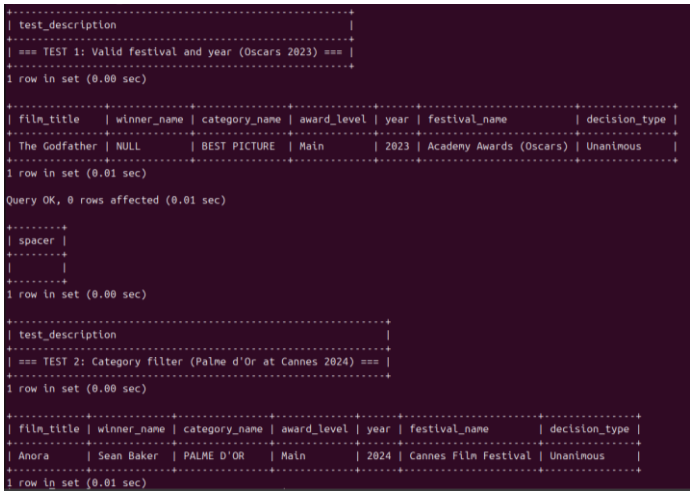
   SOURCE test_stored_procedure.sql;

   SOURCE test_triggers.sql;

   SOURCE test_views.sql;

   SOURCE test_indexes.sql;

You should see similar outputs for all:



8. **Sample Queries:** to run sample queries you can either source the files:

   SOURCE query1.1.sql;

   SOURCE query1.2.sql;

   SOURCE query1.3.sql;

   SOURCE query1.4.sql;

   SOURCE query2.1.sql;

   SOURCE query2.2.sql;

   SOURCE query2.3.sql;

   SOURCE query2.4.sql;

   or open the commands file and copy the queries you wish to execute in the MySQL terminal and see the output. Some sample outputs are as follows:

```
mysql> SELECT
    ->     ff.name as festival_name,
    ->     e.year,
    ->     e.venue,
    ->     DATEDIFF(e.end_date, e.start_date) as duration_days,
    ->     COUNT(DISTINCT n.nomination_ID) as total_nominations,
    ->     COUNT(DISTINCT CASE WHEN n.status = 'Won' THEN n.nomination_ID END) as awards_given,
    ->     COUNT(DISTINCT n.film_ID) as unique_films,
    ->     COUNT(DISTINCT n.person_ID) as unique_people
    -> FROM edition e
    -> JOIN film_festival ff ON e.festival_ID = ff.festival_ID
    -> LEFT JOIN nomination n ON e.edition_ID = n.edition_ID
    -> GROUP BY ff.name, e.year, e.venue, e.start_date, e.end_date
    -> HAVING total_nominations > 0
    -> ORDER BY ff.name, e.year DESC;
+-----------------------+------+-------------------------+--------------+-------------------+--------------+--------------+---------------+
| festival_name         | year | venue                   | duration_days | total_nominations | awards_given | unique_films | unique_people |
+-----------------------+------+-------------------------+--------------+-------------------+--------------+--------------+---------------+
| Academy Awards (Oscars) | 2025 | Dolby Theatre           |            0 |                 1 |            1 |            1 |             0 |
| Academy Awards (Oscars) | 2024 | Dolby Theatre           |            0 |                 1 |            1 |            1 |             0 |
| Academy Awards (Oscars) | 2023 | Dolby Theatre           |            0 |                 1 |            1 |            1 |             0 |
| Academy Awards (Oscars) | 2022 | Dolby Theatre           |            0 |                 1 |            1 |            1 |             0 |
| Academy Awards (Oscars) | 2021 | Union Station           |            0 |                 1 |            1 |            1 |             0 |
| Academy Awards (Oscars) | 1928 | Hollywood Roosevelt Hotel |            0 |                 2 |            2 |            2 |             2 |
| Cannes Film Festival    | 2025 | Palais des Festivals    |           11 |                 1 |            1 |            1 |             1 |
| Cannes Film Festival    | 2024 | Palais des Festivals    |           11 |                 1 |            1 |            1 |             1 |
| Cannes Film Festival    | 2019 | Palais des Festivals    |           11 |                 1 |            1 |            1 |             1 |
| Cannes Film Festival    | 1946 | Palais des Festivals    |           15 |                 1 |            1 |            1 |             1 |
+-----------------------+------+-------------------------+--------------+-------------------+--------------+--------------+---------------+
10 rows in set (0.07 sec)
```

```
mysql> SELECT
    ->     f.title,
    ->     f.release_year,
    ->     p.first_name,
    ->     p.last_name,
    ->     fc.role,
    ->     p.profession
    -> FROM film_crew fc
    -> JOIN film f ON fc.film_ID = f.film_ID
    -> JOIN person p ON fc.person_ID = p.person_ID
    -> WHERE fc.role IN ('Director', 'Producer', 'Actor')
    ->     AND f.title IS NOT NULL
    -> ORDER BY f.release_year DESC, fc.role
    -> LIMIT 20;
+-------------------------+--------------+-------------+------------+----------+------------+
| title                   | release_year | first_name  | last_name  | role     | profession |
+-------------------------+--------------+-------------+------------+----------+------------+
| It Was Just an Accident |         2025 | Jafar       | Panahi     | Director | Director   |
| Anora                   |         2024 | Sean        | Baker      | Director | Director   |
| Parasite                |         2019 | Bong        | Joon-ho    | Director | Director   |
| Inception               |         2010 | Christopher | Nolan      | Director | Director   |
| Pulp Fiction            |         1994 | Quentin     | Tarantino  | Director | Director   |
| Schindler's List        |         1993 | Steven      | Spielberg  | Director | Director   |
| The Godfather           |         1972 | Francis     | Coppola    | Director | Director   |
| Rome, Open City         |         1945 | Roberto     | Rossellini | Director | Director   |
+-------------------------+--------------+-------------+------------+----------+------------+
8 rows in set (1.98 sec)
```

Python Connectivity

This section explains how you can connect to the database from a Python environment and perform operations.

Prerequisites:

- Python3 installed
- mysql-connector-python package
- MySQL server running
- Database created and populated (from parts 2-3)

1. Before you exit your MySQL, run these commands:

ALTER USER ‘root’@’localhost’ IDENTIFIED WITH mysql_native_password BY ‘Sistersz01.1540’;

FLUSH PRIVILEGES;

Then exit your MySQL.

2. Navigate to the Python Scripts Directory:

   cd path/to/your/folder/where/you/downloaded/my/scripts

3. Run the Python Connection Script:

   python3 database_connection.py

4. Your output should be like:

